Managerial Economics (ARE) 106eh
University of California, Davis
Instructor: John H. Constantine
KEY—Practice Homework 1

Problem 1: There are three charts labeled Problem 1 to fill in at the end of this assignment. You are to solve this problem by hand. x is income and y is food expenditure.

(a) Fill in all relevant boxes for page labeled Problem 1 Chart 1 found at the end of this assignment. What are the values of b_1 and b_2? You are to calculate b_1 and b_2, and to provide a brief interpretation of what b_1 and b_2 represent. Note that you will need to fill in quite a few boxes. Recall that:

$$b_2 = \frac{\sum (x_i - \bar{x}) * (y_i - \bar{y})}{\sum (x_i - \bar{x})^2} \qquad b_1 = \bar{y} - b_2 \bar{x}$$

b_1 = intercept = 2.69 = amount of x when y = 0.

b_2 = slope = 2.69 = $\Delta y/\Delta x$

(b) On Chart 2 draw the line represented by the OLS estimators. Make sure your line goes through $\bar{x}$, $\bar{y}$. Verify your line passes through the means by plugging in the relevant numbers.

(c) What are values of (i) $\sum \hat{e}_i$ and (ii) $\sum \hat{e}_i^2$

$$\sum \hat{e}_i = 0 \qquad \sum \hat{e}_i^2 = 146.21 = \text{SSE} = \text{Sum or Squared Errors.}$$

(d) Compute the value of *σ^2-hat*. What is *σ^2-hat* measuring?

$$\hat{\sigma}^2 = \frac{\sum \hat{e}^2}{(N-K)} = \frac{146.21}{(4-2)} = 73.10 \qquad \hat{\sigma} = 8.55$$

We are more interested in, $\hat{\sigma} = 8.55$ or how the data are disbursed around the regression line.

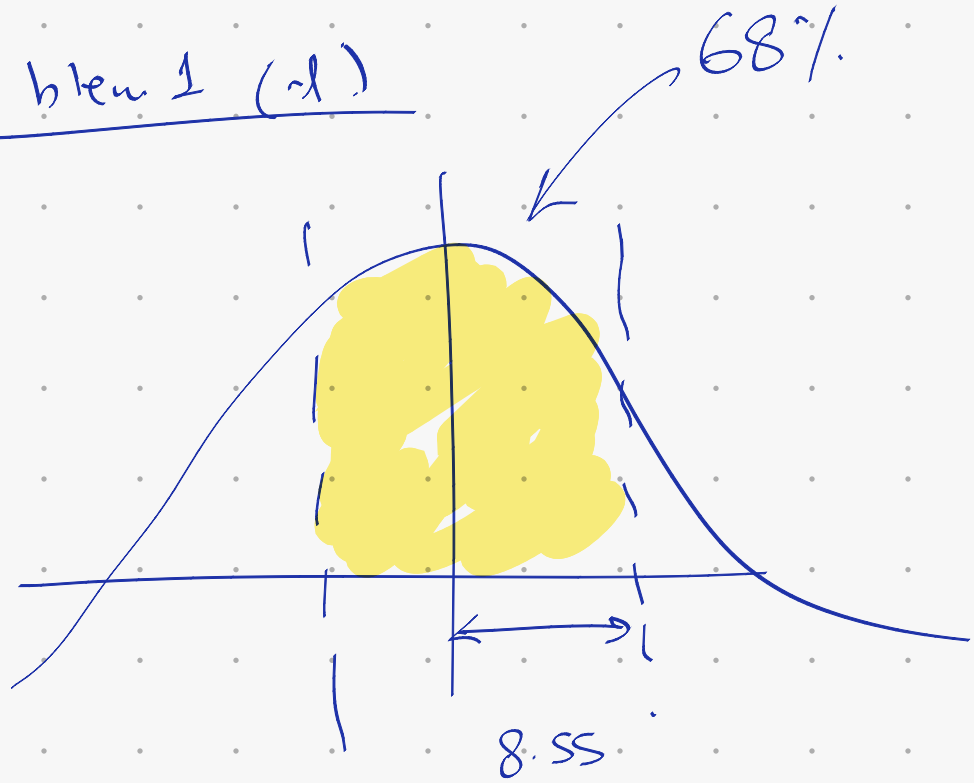
Recall, the data are centered on the regression line, but are spread out. Since we are assuming $\hat{\sigma}$ is normally distributed, that 68% of the data fall within 1 standard deviation of the regression line, etc.

(g) Compute the value of $se(\hat{b}_2)$. What is $se(\hat{b}_2)$ measuring?

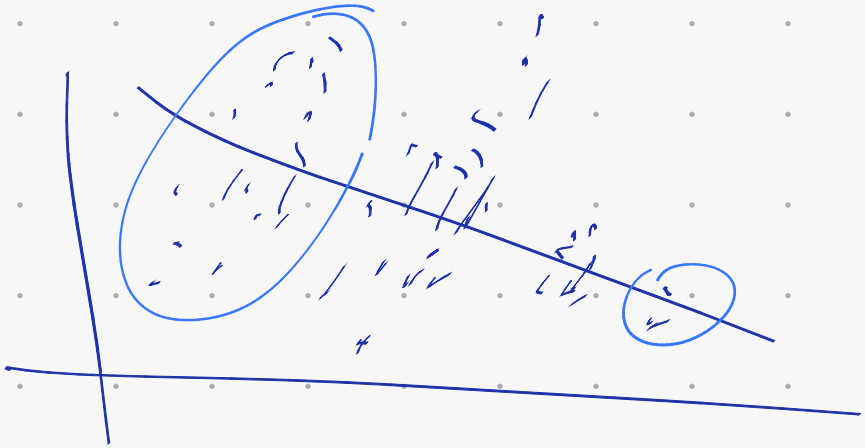
$$\hat{\sigma}_{b1}^2 = \frac{\sigma^2}{\sum (x_i - \bar{x})^2} \qquad se(\hat{b}_2) = 1.59$$

If take repeated samples, say 10 samples, we will get 10 values of b_2. These data of estimated b_2 are centered on β_2 but disbursed in a similar manner as discussed in part (f) above and in class.

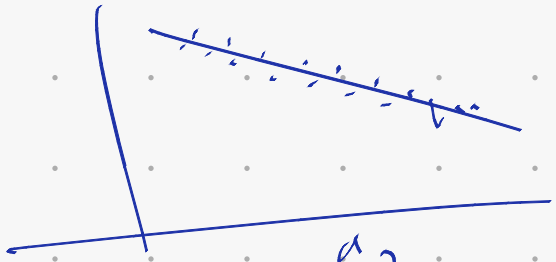
Problem 1 (d)



$$\sqrt{E[e^2|X]}$$



larger $\hat{\sigma}^2$



smaller $\hat{\sigma}^2$

Problem 2: There are two charts labeled Problem 2 to fill in at the end of this assignment. You are to solve this problem by reading the computer printout.

(a) What are the values of b_1 and b_2?

b_1 = - 195.51
b_2 = 383

(b) Provide an interpretation of b_1. How much value should we place on the results of b_1?

b_1 = the value of y when x is 0.

We should not put too much (if any) emphasis on b_1. There are no values of y in this sample that are negative. Furthermore, the data are clumped far from the data suggesting b_1 is not relevant measure here.

(c) Interpret this model's R^2? Interpret this model's "1 – R^2"

What is R-Squared? R-Squared (R^2 or the coefficient of determination) is a statistical measure in a regression model that determines the proportion of variance in the dependent variable that can be explained by the independent variable.

In econometrics, "1 - R-squared" represents the proportion of variance in the dependent variable that is not explained by the regression model; essentially, it shows the amount of variation in the data that the model cannot account for, indicating how much of the data is unexplained by the fitted regression line.

Example:
If your R-squared is 0.7, then "1 - R-squared" would be 0.3, meaning 30% of the variation in the data is not explained by the model.

Problem 3:
Explain carefully the meaning of each of the following terms (if there are acronyms, then you are to define the acronym as well):

(a) PRF.

Population regression function. It states how the population *mean* value of the dependent variable is related to one or more explanatory variables.

$$y_t = \beta_1 + \beta_2 x_t + e_t$$

(b) SRF.

Sample regression function. It is the sample counterpart of the PRF.

$$\hat{y}_t = b_1 + b_2 x_t$$

$$y_t = b_1 + b_2 x_t + \hat{e}_t$$

(c) Linear regression model.

It tells how the individual *y* are related to the explanatory variables (*x*) and the stochastic error term, $\hat{e}$, in the population as a whole. Note that this implies that the model linear in the parameters, β_1 and β_2.

(d) Residual error term, $\hat{e}_i$.

It is the sample counterpart of the stochastic error term, e_i. It is also the "residual" terms, and can be calculated as:

$$\hat{e}_i = y_i - \hat{y}_i$$

(e) Stochastic error term, e_i.

It is a proxy for all omitted or neglected variables that affect the dependent variable *y*. The individual influence of each of these variables is random and small so that on average their influence on *Y* is zero. This term is unobservable.

(f) Conditional expectation.

The expected value of *y* conditional upon a given value of *y*. It is obtained from the conditional (probability) distribution of *y*, given *x*.

(g) Estimators of regression coefficients.

The b_1 and b_2 coefficients prior to running an OLS regression. These are random variables.

(h) Estimates of regression coefficients.

The b_1 and b_2 coefficients after to running an OLS regression. These are NOT random variables; they are numbers and estimates of the true population parameters, β_1 and β_2.

<u>Problem 4</u>:
State whether the following models are linear regression models.

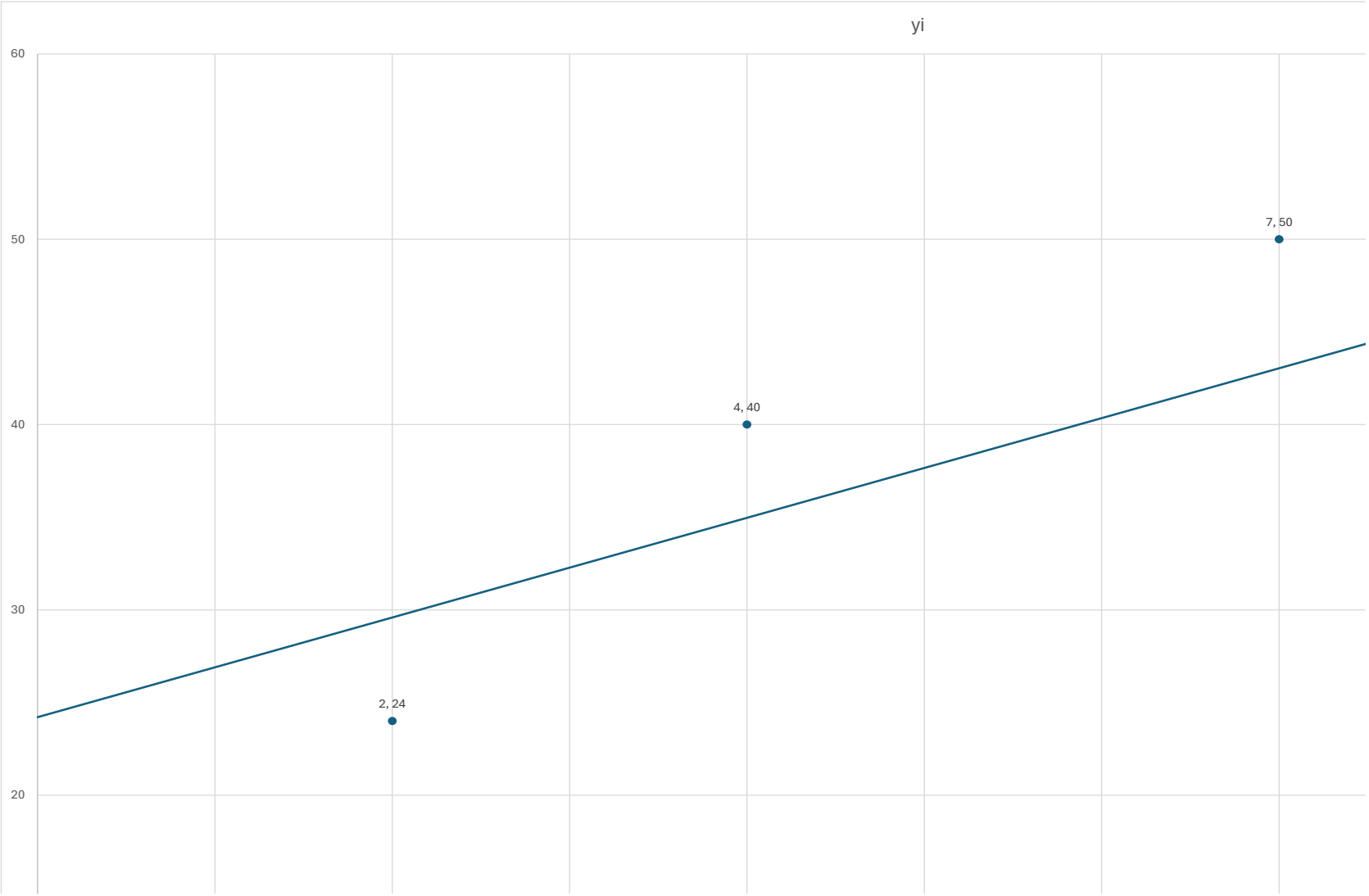
(a) $y_i = \beta_1 + \beta_2 x_i + e_i$.
(b) $y_i = \beta_1 + \beta_2(1/x_i) + e_i$.
(c) $y_i = \beta_1 + \beta_2 \ln(x_i) + e_i$.
(d) $\ln(y_i) = \beta_1 + \beta_2 x_i + e_i$.
(e) $\ln(y_i) = \beta_1 + \beta_2 \ln(x_i) + e_i$.
(f) $y_i = \beta_1 + \beta_2\beta_3 x_i + e_i$.
(g) $y_i = \beta_1 + \beta_2^2 x_i + e_i$.

(*a*)	**Yes**
(*b*)	**Yes**
(*c*)	**Yes**
(*d*)	**Yes**
(*e*)	**Yes**
(*f*)	**No**
(g)	**No**

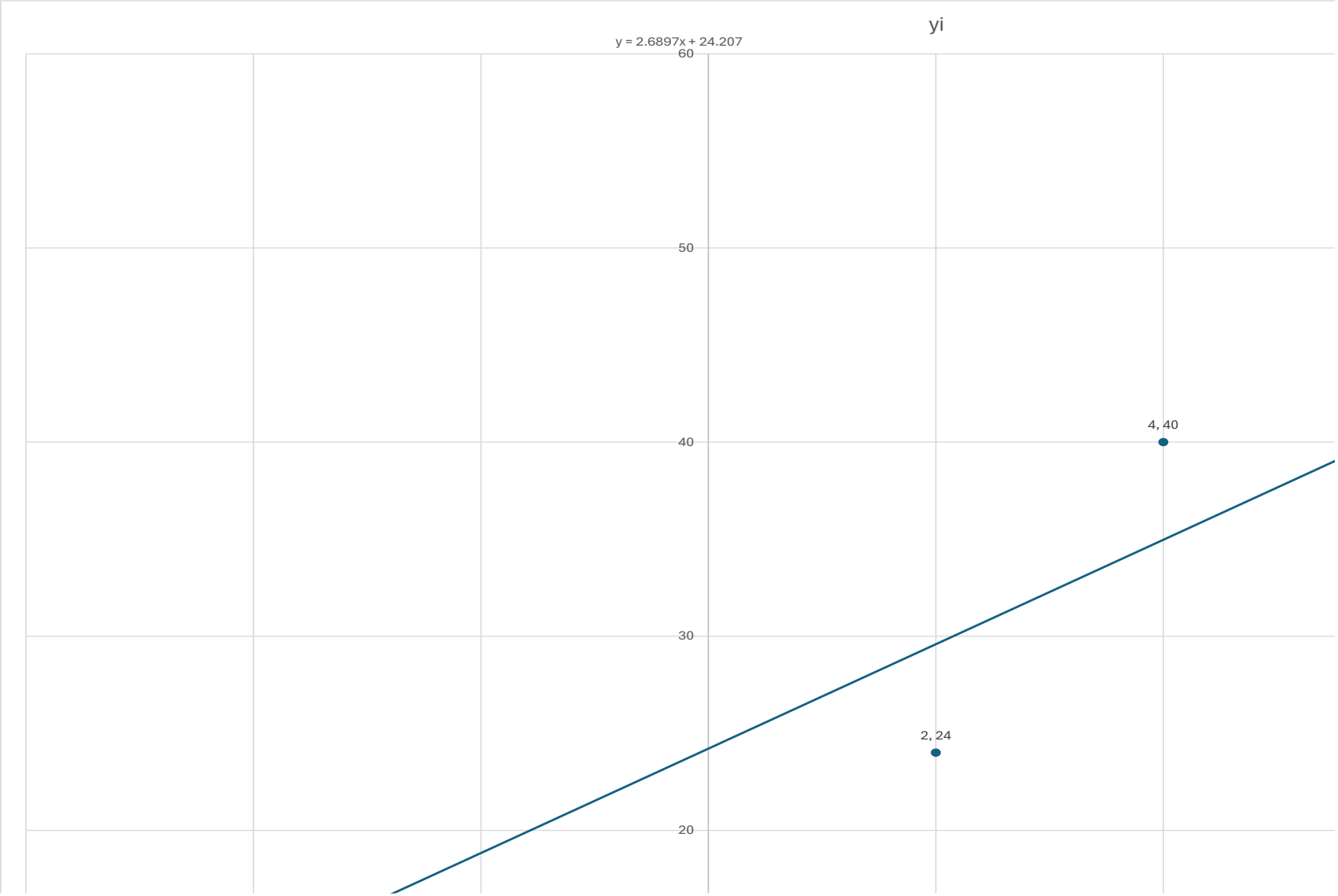
Problem 1, Chart 1:

	A	B	C	D	E	F	G	H	I	J
4			**OBS**	y_i	x_i		x_i	y_i		
5			1	24	2		2	24		
6			2	40	4		4	40		
7			3	50	7		7	50		
8			4	42	9		9	42		
9										
10	**OBS**	y_i	x_i	$y_i - \bar{y}$	$x_i - \bar{x}$	$(x_i - \bar{x}) * (y_i - \bar{y})$	$(x_i - \bar{x})^2$	$\hat{y}_i$	$\hat{e}_i$	$\hat{e}_i^2$
11	1	24	2	-15.00	-3.50	52.50	12.25	29.59	-5.59	31.21
12	2	40	4	1.00	-1.50	-1.50	2.25	34.97	5.03	25.35
13	3	50	7	11.00	1.50	16.50	2.25	43.03	6.97	48.52
14	4	42	9	3.00	3.50	10.50	12.25	48.41	-6.41	41.14
15										
16					**Sum**	**78.00**	**29.00**			**Sum ehat2**
17										**146.21**
18		**Mean**	**Mean**			**b1**	**24.21**			
19		**E(y)**	**E(x)**			**b2**	**2.69**			**σ^2-hat**
20		39.00	5.50							**73.10**
21						**var(b2)**	**2.52**			**σ**
22						**se(b2)**	**1.59**			**8.55**

Problem 1, Chart 2:



Problem 1, Chart 3:



Regression Statistics	
Multiple R	0.767662993
R Square	0.58930647
Adjusted R Square	0.383959706
Standard Error	8.550055455
Observations	4

ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
Regression	1	209.7931034	209.7931034	2.869811321	0.232337007
Residual	2	146.2068966	73.10344828		
Total	3	356			

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
Intercept	24.20689655	9.722670469	2.489737426	0.130481951	-17.62637808	66.04017119
xi	2.689655172	1.587705439	1.694051747	0.232337007	-4.141689969	9.521000314

Problem 2:spX4

The distribution of X is normal with $\mu = 50$ and $\sigma = 4$. *Although graphs are not required here they might simplify answer the question.*

(a) What is the probability that?

(i) $P(X > 43)$?

(ii) $P(X < 40.3)$?

(iii) $P(42 < X < 48)$?

(iv) $P(X \geq 57.5)$?

(v) $PX \leq 40)$ or $P(X \geq 55)$?

(i) $P(X > 43) = P(Z > -1.75) = 0.4599 + 0.5 = 0.9599$

(ii) $P(X < 42) = P(Z < -2.00) = 0.5 - 0.4772 = 0.0228$

(iii) $P(42 < X < 48) = P(-2.00 < Z < -0.50) = 0.4772 - 0.1915 = 0.2857$

(iv) $P(X > 57.5) = P(Z > 1.88) = 0.5 - 0.4699 = 0.0301$

(v) $P(X < 40) = P(Z < -2.50) = 0.5 - 0.4938 = 0.0062$

$P(X > 55) = P(Z > 1.25) = 0.5 - 0.3944 = 0.1056$

$P(X < 40) + P(X > 55) = 0.0062 + 0.1056 = 0.1118$

(b) What is the probability that 5% of the value are less than what X value?

$P(X < A) = 0.05$, so $P(A < X < 0) = 0.45$ and $P(-1.645 < Z < 0) = 0.45$

$$Z = -1.645 = \frac{X - 50}{4}$$

$\boxed{X = 50 - 1.645(4) = 43.42}$

(c) 60% of the values are between what two X values?

$P(-A < X < A) = 0.60$

$P(-0.84 < Z < 0) = 0.30$ and $P(0 < Z < 0.84) = 0.30$

$$Z = -0.84 = \frac{X_{\text{lower}} - 50}{4} \qquad Z = +0.84 = \frac{X_{\text{upper}} - 50}{4}$$

$\boxed{X_{\text{lower}} = 50 - 0.84(4) = 46.64 \quad \text{and} \quad X_{\text{upper}} = 50 + 0.84(4) = 53.36}$

(d) 85% of the values will be above what X value?

$P(X > A) = 0.85$, so $P(A < X < 0) = 0.35$

$P(-1.04 < Z < 0) = 0.35$

$$Z = -1.35 = \frac{X - 50}{4}$$

$\boxed{X = 50 - 1.35(4) = 44.60}$